

AOL_Code dan Output

2026-01-02

```
# Libraries ----
```

```
library(psych)
```

```
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
library(nortest)
```

```
library(lmtest)
```

```
## Loading required package: zoo
```

```
##
```

```
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## as.Date, as.Date.numeric
```

```
library(car)
```

```
## Loading required package: carData
```

```
##
```

```
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

```
## recode
```

```
## The following object is masked from 'package:psych':
```

```
##
```

```
## logit
```

```
library(sandwich)
```

```
# Libraries ----
```

```
# Data Preprocessing ----
```

```
# Read data
```

```
data <- read.csv("/Users/bellana/Downloads/ScreenTime vs MentalWellness.csv",  
                header = TRUE, sep = ",")
```

```
# Check missing value
```

```
sum(is.na(data))
```

```
## [1] 400
```

```
# Drop baris yang column occupationnya bukan Employed atau Self-Employed
```

```
data <- data[data$occupation == "Employed" | data$occupation == "Self Employed", ]  
head(data)
```

```
##   user_id age gender occupation work_mode screen_time_hours work_screen_hours  
## 1  U0001  33 Female   Employed   Remote           10.79             5.44  
## 2  U0002  28 Female   Employed In-person           7.40             0.37  
## 3  U0003  35 Female   Employed   Hybrid           9.78             1.09  
## 4  U0004  42 Male     Employed   Hybrid          11.13             0.56  
## 8  U0008  36 Male     Employed In-person           6.21             0.75  
## 10 U0010  34 Male     Employed   Hybrid           9.37             0.84  
##   leisure_screen_hours sleep_hours sleep_quality_1_5 stress_level_0_10  
## 1             5.35           6.63             1             9.3  
## 2             7.03           8.05             3             5.7  
## 3             8.69           6.48             1             9.1  
## 4            10.57           6.89             1            10.0  
## 8             5.46           6.78             2             5.4  
## 10            8.53           7.78             2            10.0  
##   productivity_0_100 exercise_minutes_per_week social_hours_per_week  
## 1             44.7                127             0.7  
## 2             78.0                74             2.1  
## 3             51.8                67             8.0  
## 4             37.0                 0             5.7  
## 8             87.0                51            18.7  
## 10            42.2                50            12.7  
##   mental_wellness_index_0_100 X  
## 1             9.3 NA  
## 2            56.2 NA  
## 3             3.6 NA  
## 4             0.0 NA  
## 8            54.3 NA  
## 10            10.7 NA
```

```
cat("Jumlah baris: ", nrow(data))
```

```
## Jumlah baris: 207
```

```
data <- data %>% rename("sleep_quality" = "sleep_quality_1_5",
                        "stress_level" = "stress_level_0_10",
                        "productivity" = "productivity_0_100",
                        "mental_wellness" = "mental_wellness_index_0_100")
head(data)
```

```
##   user_id age gender occupation work_mode screen_time_hours work_screen_hours
## 1  U0001  33 Female   Employed   Remote             10.79             5.44
## 2  U0002  28 Female   Employed In-person             7.40             0.37
## 3  U0003  35 Female   Employed   Hybrid             9.78             1.09
## 4  U0004  42  Male   Employed   Hybrid            11.13             0.56
## 8  U0008  36  Male   Employed In-person             6.21             0.75
## 10 U0010  34  Male   Employed   Hybrid             9.37             0.84
##   leisure_screen_hours sleep_hours sleep_quality stress_level productivity
## 1                   5.35         6.63           1           9.3         44.7
## 2                   7.03         8.05           3           5.7         78.0
## 3                   8.69         6.48           1           9.1         51.8
## 4                  10.57         6.89           1          10.0         37.0
## 8                   5.46         6.78           2           5.4         87.0
## 10                  8.53         7.78           2          10.0         42.2
##   exercise_minutes_per_week social_hours_per_week mental_wellness X
## 1                      127                0.7           9.3 NA
## 2                       74                2.1          56.2 NA
## 3                       67                8.0           3.6 NA
## 4                        0                5.7           0.0 NA
## 8                       51               18.7          54.3 NA
## 10                      50               12.7          10.7 NA
```

```
names(data)
```

```
## [1] "user_id"          "age"
## [3] "gender"           "occupation"
## [5] "work_mode"        "screen_time_hours"
## [7] "work_screen_hours" "leisure_screen_hours"
## [9] "sleep_hours"      "sleep_quality"
## [11] "stress_level"     "productivity"
## [13] "exercise_minutes_per_week" "social_hours_per_week"
## [15] "mental_wellness"  "X"
```

```
table(data$sleep_quality)
```

```
##
##   1   2   3
## 138 56 13
```

```
# Buat variable selection (backward elimination)
```

```
# Baseline 3: ----
```

```
data$sleep_quality<-as.factor(data$sleep_quality)
data$sleep_quality<-relevel(data$sleep_quality, ref = 3)
```

```
m1 <- lm(stress_level ~ age + screen_time_hours + work_screen_hours + leisure_screen_hours
        + sleep_hours + sleep_quality + productivity + exercise_minutes_per_week
        + social_hours_per_week + mental_wellness, data = data)
summary(m1)
```

```
##
## Call:
## lm(formula = stress_level ~ age + screen_time_hours + work_screen_hours +
##     leisure_screen_hours + sleep_hours + sleep_quality + productivity +
##     exercise_minutes_per_week + social_hours_per_week + mental_wellness,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.30312 -0.48831 -0.00857  0.47675  2.15707
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    13.2067488   1.0661479   12.387 < 2e-16 ***
## age              0.0018302   0.0071041    0.258 0.796971
## screen_time_hours  0.0505728   0.0387536    1.305 0.193428
## work_screen_hours -0.0426555   0.0365795   -1.166 0.244989
## leisure_screen_hours      NA         NA         NA      NA
## sleep_hours      -0.0604333   0.0867520   -0.697 0.486865
## sleep_quality1    -1.7807439   0.3357996   -5.303 3.06e-07 ***
## sleep_quality2    -1.0497293   0.2712178   -3.870 0.000148 ***
## productivity      -0.0347602   0.0090774   -3.829 0.000173 ***
## exercise_minutes_per_week  0.0016280   0.0008222    1.980 0.049092 *
## social_hours_per_week -0.0112766   0.0115408   -0.977 0.329722
## mental_wellness    -0.0858984   0.0079917  -10.748 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7625 on 196 degrees of freedom
## Multiple R-squared:  0.8592, Adjusted R-squared:  0.852
## F-statistic: 119.6 on 10 and 196 DF,  p-value: < 2.2e-16
```

```
m2 <- lm(stress_level ~ screen_time_hours + work_screen_hours + leisure_screen_hours
        + sleep_hours + sleep_quality + productivity + exercise_minutes_per_week
        + social_hours_per_week + mental_wellness, data = data)
summary(m2)
```

```
##
## Call:
## lm(formula = stress_level ~ screen_time_hours + work_screen_hours +
##     leisure_screen_hours + sleep_hours + sleep_quality + productivity +
##     exercise_minutes_per_week + social_hours_per_week + mental_wellness,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.30430 -0.47480 -0.01006  0.48363  2.14594
```

```
##
## Coefficients: (1 not defined because of singularities)
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      13.2713873   1.0337448   12.838 < 2e-16 ***
## screen_time_hours    0.0498279   0.0385539    1.292 0.197725
## work_screen_hours   -0.0420837   0.0364255   -1.155 0.249352
## leisure_screen_hours      NA         NA         NA      NA
## sleep_hours        -0.0606481   0.0865422   -0.701 0.484260
## sleep_quality1     -1.7776178   0.3347841   -5.310 2.95e-07 ***
## sleep_quality2     -1.0398390   0.2678499   -3.882 0.000141 ***
## productivity       -0.0349555   0.0090242   -3.874 0.000146 ***
## exercise_minutes_per_week 0.0016305 0.0008202    1.988 0.048194 *
## social_hours_per_week -0.0111046 0.0114942   -0.966 0.335173
## mental_wellness     -0.0858840 0.0079725  -10.773 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7607 on 197 degrees of freedom
## Multiple R-squared:  0.8592, Adjusted R-squared:  0.8527
## F-statistic: 133.5 on 9 and 197 DF,  p-value: < 2.2e-16

m3 <- lm(stress_level ~ screen_time_hours + work_screen_hours + leisure_screen_hours
        + sleep_quality + productivity + exercise_minutes_per_week + social_hours_per_week
        + mental_wellness, data = data)
summary(m3)

##
## Call:
## lm(formula = stress_level ~ screen_time_hours + work_screen_hours +
##     leisure_screen_hours + sleep_quality + productivity + exercise_minutes_per_week +
##     social_hours_per_week + mental_wellness, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.26813 -0.48329 -0.00661  0.47737  2.11167
##
## Coefficients: (1 not defined because of singularities)
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      12.8013798   0.7856367   16.294 < 2e-16 ***
## screen_time_hours    0.0519068   0.0383902    1.352 0.177890
## work_screen_hours   -0.0428142   0.0363637   -1.177 0.240455
## leisure_screen_hours      NA         NA         NA      NA
## sleep_quality1     -1.7348678   0.3287562   -5.277 3.43e-07 ***
## sleep_quality2     -1.0342285   0.2673859   -3.868 0.000149 ***
## productivity       -0.0346168   0.0089997   -3.846 0.000162 ***
## exercise_minutes_per_week 0.0015738 0.0008151    1.931 0.054932 .
## social_hours_per_week -0.0107617 0.0114690   -0.938 0.349216
## mental_wellness     -0.0867291 0.0078707  -11.019 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7597 on 198 degrees of freedom
## Multiple R-squared:  0.8588, Adjusted R-squared:  0.8531
## F-statistic: 150.6 on 8 and 198 DF,  p-value: < 2.2e-16
```

```
m4 <- lm(stress_level ~ screen_time_hours + work_screen_hours + leisure_screen_hours
          + sleep_quality + productivity + exercise_minutes_per_week + mental_wellness,
          data = data)
summary(m4)
```

```
##
## Call:
## lm(formula = stress_level ~ screen_time_hours + work_screen_hours +
##     leisure_screen_hours + sleep_quality + productivity + exercise_minutes_per_week +
##     mental_wellness, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.27363 -0.48715 -0.00998  0.49821  2.05285
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    12.6394295   0.7662139   16.496 < 2e-16 ***
## screen_time_hours    0.0614989   0.0369930    1.662 0.097997 .
## work_screen_hours   -0.0457836   0.0362149   -1.264 0.207630
## leisure_screen_hours      NA         NA         NA      NA
## sleep_quality1     -1.7203715   0.3282943   -5.240 4.07e-07 ***
## sleep_quality2     -1.0201364   0.2668836   -3.822 0.000177 ***
## productivity       -0.0352985   0.0089676   -3.936 0.000114 ***
## exercise_minutes_per_week  0.0015823   0.0008148    1.942 0.053556 .
## mental_wellness    -0.0858489   0.0078122  -10.989 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7595 on 199 degrees of freedom
## Multiple R-squared:  0.8582, Adjusted R-squared:  0.8532
## F-statistic: 172 on 7 and 199 DF, p-value: < 2.2e-16
```

```
m5 <- lm(stress_level ~ screen_time_hours + leisure_screen_hours + sleep_quality
          + productivity + exercise_minutes_per_week + mental_wellness, data = data)
summary(m5)
```

```
##
## Call:
## lm(formula = stress_level ~ screen_time_hours + leisure_screen_hours +
##     sleep_quality + productivity + exercise_minutes_per_week +
##     mental_wellness, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.27363 -0.48715 -0.00998  0.49821  2.05285
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    12.6394295   0.7662139   16.496 < 2e-16 ***
## screen_time_hours    0.0157153   0.0448376    0.350 0.726338
## leisure_screen_hours    0.0457836   0.0362149    1.264 0.207630
```

```
## sleep_quality1          -1.7203715  0.3282943  -5.240 4.07e-07 ***
## sleep_quality2          -1.0201364  0.2668836  -3.822 0.000177 ***
## productivity            -0.0352985  0.0089676  -3.936 0.000114 ***
## exercise_minutes_per_week 0.0015823  0.0008148   1.942 0.053556 .
## mental_wellness         -0.0858489  0.0078122 -10.989 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7595 on 199 degrees of freedom
## Multiple R-squared:  0.8582, Adjusted R-squared:  0.8532
## F-statistic: 172 on 7 and 199 DF, p-value: < 2.2e-16
```

```
m6 <- lm(stress_level ~ leisure_screen_hours + sleep_quality + productivity
          + exercise_minutes_per_week + mental_wellness, data = data)
summary(m6)
```

```
##
## Call:
## lm(formula = stress_level ~ leisure_screen_hours + sleep_quality +
##      productivity + exercise_minutes_per_week + mental_wellness,
##      data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.27314 -0.48733 -0.01333  0.51129  2.05999
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    12.8144587   0.5798491   22.100 < 2e-16 ***
## leisure_screen_hours  0.0534186   0.0288673    1.850  0.0657 .
## sleep_quality1      -1.7579300   0.3096319   -5.677 4.76e-08 ***
## sleep_quality2      -1.0398628   0.2603090   -3.995 9.10e-05 ***
## productivity       -0.0361189   0.0086378   -4.182 4.33e-05 ***
## exercise_minutes_per_week 0.0015625  0.0008111    1.926  0.0555 .
## mental_wellness     -0.0864875   0.0075801  -11.410 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7578 on 200 degrees of freedom
## Multiple R-squared:  0.8581, Adjusted R-squared:  0.8538
## F-statistic: 201.6 on 6 and 200 DF, p-value: < 2.2e-16
```

```
m7 <- lm(stress_level ~ sleep_quality + productivity + exercise_minutes_per_week
          + mental_wellness, data = data)
summary(m7)
```

```
##
## Call:
## lm(formula = stress_level ~ sleep_quality + productivity + exercise_minutes_per_week +
##      mental_wellness, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -2.22616 -0.46397 0.02472 0.46898 2.16467
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    13.4939214   0.4514780   29.888 < 2e-16 ***
## sleep_quality1  -1.8887955   0.3032601   -6.228 2.71e-09 ***
## sleep_quality2  -1.1245395   0.2577962   -4.362 2.06e-05 ***
## productivity    -0.0385606   0.0085877   -4.490 1.20e-05 ***
## exercise_minutes_per_week 0.0015577 0.0008159    1.909 0.0577 .
## mental_wellness -0.0893562   0.0074645  -11.971 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7624 on 201 degrees of freedom
## Multiple R-squared:  0.8557, Adjusted R-squared:  0.8521
## F-statistic: 238.3 on 5 and 201 DF,  p-value: < 2.2e-16
```

```
m8 <- lm(stress_level ~ sleep_quality + productivity + mental_wellness, data = data)
summary(m8)
```

```
##
## Call:
## lm(formula = stress_level ~ sleep_quality + productivity + mental_wellness,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.37531 -0.41469  0.00536  0.48327  2.24121
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    13.668548   0.444999   30.716 < 2e-16 ***
## sleep_quality1 -1.794164   0.301133   -5.958 1.12e-08 ***
## sleep_quality2 -1.067851   0.257751   -4.143 5.04e-05 ***
## productivity   -0.041679   0.008486   -4.912 1.86e-06 ***
## mental_wellness -0.085346   0.007210  -11.838 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7674 on 202 degrees of freedom
## Multiple R-squared:  0.8531, Adjusted R-squared:  0.8501
## F-statistic: 293.2 on 4 and 202 DF,  p-value: < 2.2e-16
```

```
data <- data %>% select(-c(user_id, age, gender, occupation, work_mode, screen_time_hours,
                           work_screen_hours, leisure_screen_hours, sleep_hours,
                           exercise_minutes_per_week, social_hours_per_week, X))

head(data)
```

```
##      sleep_quality stress_level productivity mental_wellness
## 1                1          9.3          44.7             9.3
## 2                3           5.7          78.0            56.2
## 3                1           9.1          51.8             3.6
```

```
## 4          1          10.0          37.0          0.0
## 8          2           5.4          87.0          54.3
## 10         2          10.0          42.2          10.7
```

```
model <- lm(stress_level ~ ., data = data)
summary(model)
```

```
##
## Call:
## lm(formula = stress_level ~ ., data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.37531 -0.41469  0.00536  0.48327  2.24121
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   13.668548   0.444999   30.716 < 2e-16 ***
## sleep_quality1 -1.794164   0.301133   -5.958 1.12e-08 ***
## sleep_quality2 -1.067851   0.257751   -4.143 5.04e-05 ***
## productivity   -0.041679   0.008486   -4.912 1.86e-06 ***
## mental_wellness -0.085346   0.007210  -11.838 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7674 on 202 degrees of freedom
## Multiple R-squared:  0.8531, Adjusted R-squared:  0.8501
## F-statistic: 293.2 on 4 and 202 DF, p-value: < 2.2e-16
```

```
std_resid <- rstandard(model)
outlier_3sd <- which(abs(std_resid) > 3)
outlier_3sd
```

```
## 227
## 116
```

```
data$.std_resid <- rstandard(model)
data_clean <- subset(data, abs(.std_resid) <= 3)
summary(model_clean <- lm(
  stress_level ~ sleep_quality + mental_wellness + productivity,
  data = data_clean
))
```

```
##
## Call:
## lm(formula = stress_level ~ sleep_quality + mental_wellness +
##      productivity, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.95772 -0.43762 -0.00629  0.47912  2.23004
##
```

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   13.594967   0.435764  31.198 < 2e-16 ***
## sleep_quality1 -1.769222   0.294576  -6.006 8.77e-09 ***
## sleep_quality2 -1.013818   0.252615  -4.013 8.45e-05 ***
## mental_wellness -0.085425   0.007050 -12.117 < 2e-16 ***
## productivity  -0.040647   0.008305  -4.895 2.02e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7504 on 201 degrees of freedom
## Multiple R-squared:  0.8562, Adjusted R-squared:  0.8534
## F-statistic: 299.2 on 4 and 201 DF,  p-value: < 2.2e-16
```

```
h <- hatvalues(model_clean)
summary(h)
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 0.007249 0.011451 0.018333 0.024272 0.027151 0.148282
```

```
idx = which(h < 3 * (length(coef(model)) / nobs(model)))
model_trim = lm(stress_level ~ sleep_quality + mental_wellness + productivity,
                 data = model_clean$model[idx, ])
summary(model_trim)
```

```
##
## Call:
## lm(formula = stress_level ~ sleep_quality + mental_wellness +
##     productivity, data = model_clean$model[idx, ])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.95672 -0.38363 -0.02552  0.48264  2.23261
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   11.701447   0.348674  33.560 < 2e-16 ***
## sleep_quality2  0.769684   0.158998   4.841 2.68e-06 ***
## mental_wellness -0.088110   0.007272 -12.116 < 2e-16 ***
## productivity   -0.037449   0.008378  -4.470 1.35e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7443 on 189 degrees of freedom
## Multiple R-squared:  0.8311, Adjusted R-squared:  0.8284
## F-statistic: 310 on 3 and 189 DF,  p-value: < 2.2e-16
```

```
# Uji Error: ----
# Uji Normalitas
e <- resid(model_trim)
lillie.test(e)
```

```
##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: e
## D = 0.0588, p-value = 0.1062
```

Kesimpulan: Error berdistribusi normal ($0.1062 > 0.05 \rightarrow$ gagal tolak H_0)

```
# Uji Heterokedastisitas
bptest(model_trim)
```

```
##
## studentized Breusch-Pagan test
##
## data: model_trim
## BP = 11.417, df = 3, p-value = 0.00967
```

Kesimpulan: Uji heteroskedastisitas tidak terpenuhi ($0.0096 < 0.05 \rightarrow$ tolak H_0)

```
# Uji Non-autokorelasi
dwtest(model_trim)
```

```
##
## Durbin-Watson test
##
## data: model_trim
## DW = 2.0733, p-value = 0.6939
## alternative hypothesis: true autocorrelation is greater than 0
```

Kesimpulan: Tidak Ada autokorelasi dalam residual ($0.6939 > 0.05 \rightarrow$ gagal tolak H_0)

```
# Uji Multikolinear
vif(model_trim)
```

```
## sleep_quality mental_wellness productivity
## 1.794449 4.782276 3.859980
```

Kesimpulan: Tidak ada multikolinearitas

```
data_gls <- data_clean[idx, ]
data_gls <- na.omit(data_gls)
```

```
str(data_gls)
```

```
## 'data.frame': 193 obs. of 5 variables:
## $ sleep_quality : Factor w/ 3 levels "3","1","2": 2 2 2 3 3 3 2 2 2 2 ...
## $ stress_level : num 9.3 9.1 10 5.4 10 6.3 10 10 10 3.7 ...
## $ productivity : num 44.7 51.8 37 87 42.2 70.7 53.2 37.6 60.3 77.3 ...
## $ mental_wellness: num 9.3 3.6 0 54.3 10.7 40.9 10.7 0 10.5 48.7 ...
## $ .std_resid : num 0.1078 -0.4053 -0.4357 1.4107 0.0948 ...
```

```
library(nlme)
```

```
##
```

```
## Attaching package: 'nlme'
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

```
## collapse
```

```
gls_model <- gls(  
  stress_level ~ sleep_quality + mental_wellness + productivity,  
  data = data_gls,  
  weights = varPower(form = ~ fitted(.))  
)
```

```
summary(gls_model)
```

```
## Generalized least squares fit by REML
```

```
## Model: stress_level ~ sleep_quality + mental_wellness + productivity
```

```
## Data: data_gls
```

```
## AIC BIC logLik
```

```
## 452.3171 471.7676 -220.1586
```

```
##
```

```
## Variance function:
```

```
## Structure: Power of variance covariate
```

```
## Formula: ~fitted(.)
```

```
## Parameter estimates:
```

```
## power
```

```
## -1.094839
```

```
##
```

```
## Coefficients:
```

```
## Value Std.Error t-value p-value
```

```
## (Intercept) 11.442011 0.3171352 36.07928 0
```

```
## sleep_quality2 0.734081 0.1604268 4.57580 0
```

```
## mental_wellness -0.082404 0.0071678 -11.49651 0
```

```
## productivity -0.033583 0.0076469 -4.39175 0
```

```
##
```

```
## Correlation:
```

```
## (Intr) slp_q2 mntl_w
```

```
## sleep_quality2 -0.027
```

```
## mental_wellness 0.604 -0.483
```

```
## productivity -0.976 0.048 -0.711
```

```
##
```

```
## Standardized residuals:
```

```
## Min Q1 Med Q3 Max
```

```
## -2.96122755 -0.59208166 -0.02317375 0.61548098 2.54574856
```

```
##
```

```
## Residual standard error: 7.299846
```

```
## Degrees of freedom: 193 total; 189 residual
```

```
res_gls <- resid(gls_model, type = "normalized")
fit_gls <- fitted(gls_model)
```

```
bptest(res_gls ~ fit_gls)
```

```
##
## studentized Breusch-Pagan test
##
## data:  res_gls ~ fit_gls
## BP = 0.033216, df = 1, p-value = 0.8554
```